OMER BURAK DEMIREL, Ph.D.

202 4th St SW, Rochester, MN 55902

(612) 475-6188 obdemirel@gmail.com burak.demirel@philips.com

SUMMARY

MRI Clinical Scientist at Philips Healthcare, supporting Mayo Clinic, and researcher with expertise in magnetic resonance imaging, computer vision, and deep learning for image reconstruction, image enhancement, and quantitative analysis. Experienced in translating advanced imaging algorithms from research to clinical practice.

PROFESSIONAL EXPERIENCE

MRI Clinical Scientist Nov 2024 – Present

Philips Healthcare, Rochester, MN, USA

- Support clinical research and product development in cardiac and interventional MRI.
- Collaborate with Mayo Clinic and Philips R&D teams to develop advanced algorithms for image reconstruction and post-processing, and to optimize MRI scanner workflows and imaging sequences.

Postdoctoral Research Fellow Aug 2023 – Jul 2024

Beth Israel Deaconess Medical Center, Harvard University, Boston, MA, USA

- Developed deep learning algorithms for accelerated cine and 3D late gadolinium enhancement cardiac MRI.
- Deployed inline reconstruction algorithms currently used in clinical imaging.

Research Assistant 2017 – 2023

Center for Magnetic Resonance Research, University of Minnesota, Minneapolis, MN

- Designed self-supervised deep learning methods for high-resolution cardiac MRI and fMRI.
- Contributed to the Human Connectome Project through novel reconstruction pipelines.

Research Assistant 2012 – 2017

National Magnetic Resonance Research Center, Bilkent University, Turkey

- Built a custom magnetic particle imaging scanner and performed nerve stimulation safety studies.

EDUCATION

University of Minnesota, Minneapolis, MN, USA 2017 – 2023

Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Mehmet Akçakaya)

Physics-Driven Deep Learning Techniques for High-Resolution MRI

Bilkent University, Ankara, Turkey 2010 – 2017

M.Sc. and B.Sc. in Electrical and Electronics Engineering

SELECTED PUBLICATIONS

- **O. B. Demirel**, S. Yoon, F. Ghanbari, C. W. Hoeger, C. W. Tsao, T. Wallace, P. Pierce, S. Johnson, K. Arcand, J. Street, J. Rodriguez, K. Chow, W. J. Manning, R. Nezafat, "Accelerated Late Gadolinium Enhancement CMR with Generative AI," Submitted to *Journal of Cardiovascular Magnetic Resonance*, 2024.
- **O. B. Demirel**, F. Ghanbari, M. A. Morales, P. Pierce, S. Johnson, J. Rodriguez, R. Nezafat, "Accelerated Cardiac Cine with Spatio-Coil Regularized Deep Learning Reconstruction," Submitted to *Magnetic Resonance in Medicine*, 2024.
- **O. B. Demirel**, B. Yaman, C. Shenoy, S. Moeller, S. Weingärtner, M. Akçakaya, "Signal Intensity Informed Multi-Coil Encoding Operator for Physics-Guided Deep Learning Reconstruction of Highly Accelerated Myocardial Perfusion CMR," *Magnetic Resonance in Medicine*, 2022: 1–14.
- **O. B. Demirel**, S. Weingärtner, S. Moeller, M. Akçakaya, "Improved Simultaneous Multislice Cardiac MRI Using Readout-Concatenated k-Space SPIRiT," *Magnetic Resonance in Medicine*, 85 (2021): 3036–3048.
- **O. B. Demirel**, T. Kilic, T. Cukur, E. U. Saritas, "Anatomical Measurements Correlate with Individual Magnetostimulation Thresholds for kHz-range Homogeneous Magnetic Fields," *Medical Physics*, 47(4) (2020): 1836–1844.

PATENTS

- "Noise Suppressed Deep Learning Reconstruction," US Patent Pending.
- "Parallel MRI Reconstruction with Multi-k-space Interpolation," US Patent Pending.

HONORS & AWARDS

- American Heart Association Predoctoral Fellowship (0.12% rank), 2020–2021
- ISMRM Summa Cum Laude Abstracts (2020, 2021); Magna Cum Laude Abstract (2024)
- NSF Travel Award, 27th EUSIPCO, 2019
- Gary H. Glover and Bruce J. Bergman Graduate Fellowships, Univ. of Minnesota
- High Honor Student, Bilkent University; Merit Scholarship 100%

TECHNICAL SKILLS

Python, PyTorch, MATLAB, LaTeX, C++, Java, VHDL, Adobe Illustrator, Deep learning, MRI reconstruction, quantitative imaging, image enhancement, computer vision.